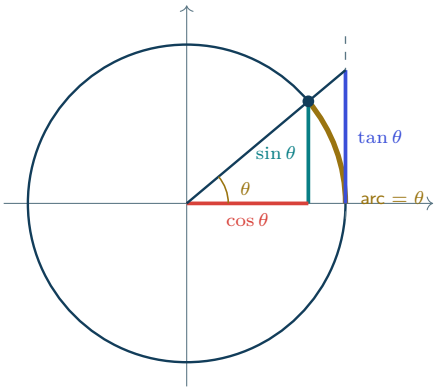
FUNCTIONS LADDER 1 OF 10

Circular functions

Sine and cosine are the coordinates of a point going round a unit circle, and tangent is the slope of the line to that point.

1 / THE RULE AND THE PICTURE

On the unit circle the point at angle θ is $(\cos\theta, \sin\theta)$, and θ is the arc length in radians



$\theta = 40° = 0.698$ rad: $\cos = 0.766$, $\sin = 0.643$, $\tan = 0.839$

A full turn is 2π radians, so $180° = \pi$ and one radian is $57.3°$.
The point lies on the circle $x^2 + y^2 = 1$, which gives the Pythagorean identity, and tangent is the ratio of the two coordinates:

$$\cos^2\theta + \sin^2\theta = 1, \qquad \tan\theta = \frac{\sin\theta}{\cos\theta}$$

2 / THE EXACT VALUES / ANGLES WITH SIMPLE SURDS

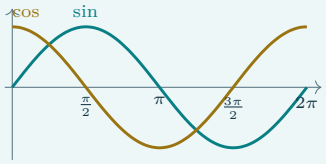
Three right triangles give every entry in the table

angle	sin	cos	tan
0	0	1	0
$\frac{\pi}{6} = 30°$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{3}}$
$\frac{\pi}{4} = 45°$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1
$\frac{\pi}{3} = 60°$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$
$\frac{\pi}{2} = 90°$	1	0	undefined

The $45°$ values come from the isosceles right triangle with sides $1, 1, \sqrt{2}$, and the $30°$ and $60°$ values from half an equilateral triangle with sides $1, \sqrt{3}, 2$.

3 / THE GRAPHS / PERIOD AND PHASE

Sine and cosine repeat every 2π, and cosine is sine shifted by a quarter turn



Both oscillate between -1 and 1, so $A\sin(\omega x + \phi)$ has amplitude A, period $2\pi/\omega$ and a shift ϕ. Since $\cos x = \sin(x + \frac{\pi}{2})$, the cosine curve is the sine curve moved left by $\frac{\pi}{2}$, and tangent repeats with the shorter period π, with vertical asymptotes where $\cos = 0$.

4 / IDENTITIES / COMBINING ANGLES

The addition formulas generate the double-angle formulas

$$\sin(A+B) = \sin A\cos B + \cos A\sin B, \qquad \cos(A+B) = \cos A\cos B - \sin A\sin B.$$

Setting $B = A$ gives $\sin 2A = 2\sin A\cos A$ and $\cos 2A = \cos^2 A - \sin^2 A = 2\cos^2 A - 1 = 1 - 2\sin^2 A$. For example $\sin 75° = \sin(45° + 30°) = \frac{\sqrt{2}}{2}\cdot\frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2}\cdot\frac{1}{2} = \frac{\sqrt{6}+\sqrt{2}}{4} = 0.9659$. Also $\sin(\pi - \theta) = \sin\theta$, $\cos(-\theta) = \cos\theta$ and $\sin(-\theta) = -\sin\theta$.

5 / INVERSES AND EQUATIONS

$\sin x = \frac{1}{2}$ has infinitely many solutions, and the arcsine picks one

The inverse $\arcsin$ returns values in $[-\frac{\pi}{2}, \frac{\pi}{2}]$, so $\arcsin\frac{1}{2} = \frac{\pi}{6}$, and the full solution set of $\sin x = \frac{1}{2}$ is $x = \frac{\pi}{6} + 2\pi n$ or $x = \frac{5\pi}{6} + 2\pi n$, the second coming from $\sin(\pi - \theta) = \sin\theta$. Similarly $\arccos$ returns $[0, \pi]$ and $\arctan$ returns $(-\frac{\pi}{2}, \frac{\pi}{2})$.

ASIDE / DEGREES OR RADIANS

Calculus formulas such as $\frac{d}{dx}\sin x = \cos x$ hold only when x is in radians, which is why radians are the natural unit.

LINKS TO OTHER SHEETS

The surds are developed in *The unit circle, the Ailles rectangle and the sine of* $15°$; the next sheet is *Hyperbolic functions*.

Convention. θ is in radians unless marked in degrees; $\sin, \cos, \tan$ are the circular functions; $\arcsin$ etc. are principal inverses.